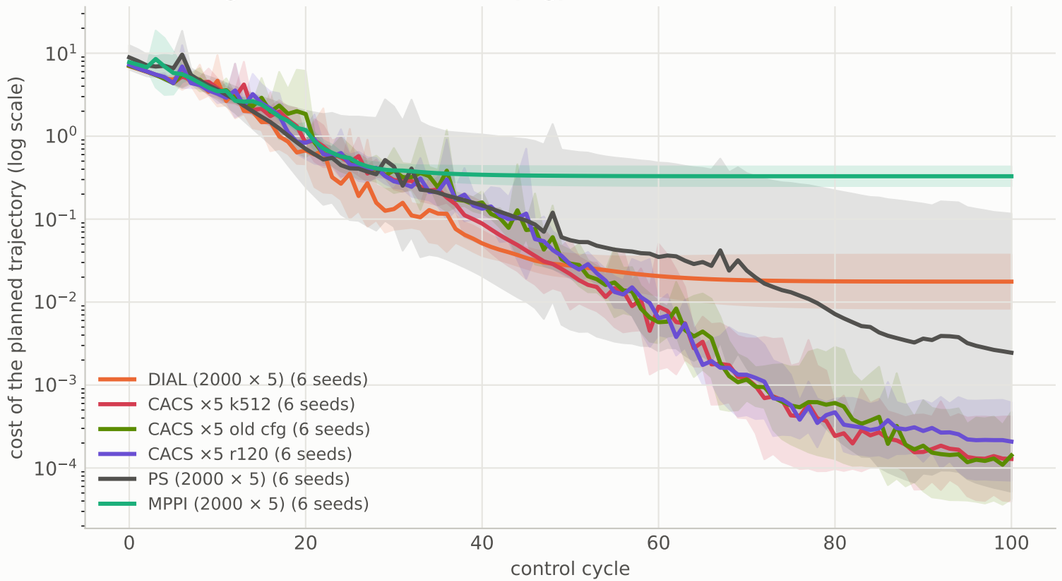
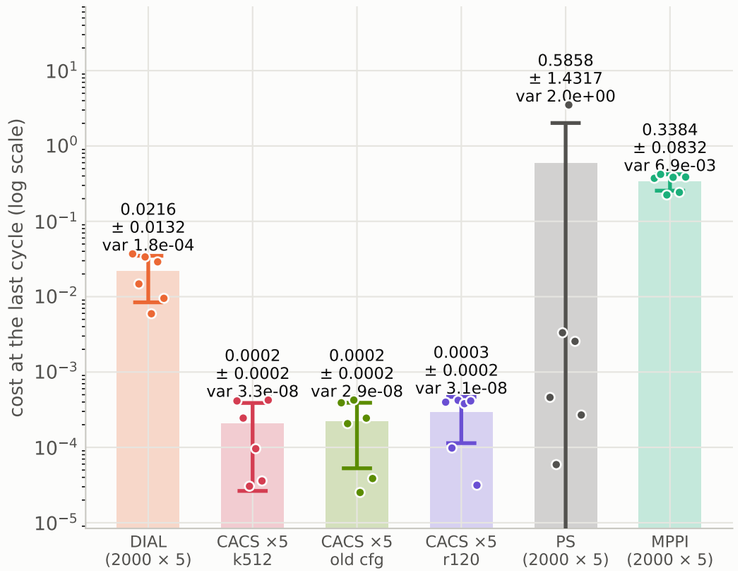
PushT closed loop: GlobalMPPI (KSOS ADMM-cuda + MPPI σ 0.1) vs PS / MPPI / DIAL

start: block (0.1, 0.1, 1.3), pusher (0, 0); 100 cycles at 50 Hz; seeds 0–5 for every method. GlobalMPPI: CACS ×5 + warm start, one cost-aware chordal sampling pass (1000 samples) per restart in shrinking boxes (decay 0.8), KSOS on ADMM-cuda, then MPPI 1000 × 5, σ 0.1. CACS: h 0.7, 200 initial Sobol, batch 768, cliques < 20, no replacements unless stated; k512 = reuse 40, k 512; r120 = reuse 120, k 512; old cfg = h 1.2, k 128, 40 replacements. PS / MPPI / DIAL: pusht.py settings with 2000 × 5 (≈ 10,000 rollouts). Final cost = cost at cycle 100. Time statistics leave out cycle 0 (JIT); std uses ddof = 1

Cost curve: geometric mean ± 1 std (log) over seeds



Final cost: mean ± std over seeds



Planning time per cycle: mean ± std over seeds

